

Meher Chaitanya Pindiprolu, PhD

MACHINE LEARNING RESEARCHER | Efficient AI | LLM Optimization | Graph Machine Learning

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Profile

Machine Learning Researcher with a PhD in Network Science from ETH Zürich, specializing in compute- and memory-efficient AI. Leads research on LLM pretraining efficiency, Tokenized Graph Transformers, long-range information flow, and scalable graph machine learning using optimization and numerical linear algebra. Combines research leadership with experience translating novel algorithms into production AI systems with measurable impact.

Core Expertise & Skills

AI:

Efficient LLM Training | KV-Cache Optimization | Graph Machine Learning | Long-Range Information Flow | Graph Transformers | Graph Neural Networks | Recommender Systems

Methods:

Numerical Linear Algebra | Optimization | Spectral Methods | Graph Algorithms | Randomized Algorithms

Programming & Frameworks:

Python | C++ | C | PyTorch | PyTorch Geometric | DGL | scikit-learn | NetworkX

Work Experience

KTH Royal Institute of Technology

Stockholm, Sweden

WALLENBERG AI POSTDOCTORAL RESEARCHER

2025– Present

- Define and lead research on efficient AI, originating projects in LLM pretraining efficiency, tokenized graph transformers, long-range information flow, and numerical methods for scalable graph learning.
- Drive end-to-end research from problem formulation to publication, including sparse Graph Transformers with up to 11% gains and large-scale LLM optimization experiments.
- Provide scientific direction to PhD and Master’s researchers, shaping methodology, experimental design, interpretation, and cross-university collaborations.

Coverstar

Stockholm, Sweden

AI CONSULTANT – RECOMMENDER SYSTEM ARCHITECT (part-time)

2025-2026

- Architected Coverstar’s next-generation “For You” recommender, combining graph retrieval, session Transformers, cold-start methods, LLM-assisted content understanding, ranking, diversity, and safety.
- Identified that short-term engagement optimization reduced diversity and long-term retention, motivating a multi-objective ranking strategy balancing relevance, diversity, retention, latency, and scalability.
- Delivered a 14.3 percentage-point improvement in 30-day retention in A/B testing versus the NVIDIA Merlin production baseline.

ETH Zürich

Zürich, Switzerland

PHD RESEARCHER

2020– 2024

- Introduced the first node-embedding framework connecting s–t min-cuts with contagion dynamics, improving node classification accuracy by up to 12%; also developed graph-Laplacian methods for link recommendation with provable guarantees.
- Designed the first scalable online algorithm for local clustering in multiplex networks, enabling efficient recovery of local structure without global graph exploration.
- Led SNSF-funded research from problem formulation through publication and mentored two Master’s students on methodology, experiments, and interpretation.

IIIT Hyderabad	Hyderabad, India
HIGH PERFORMANCE COMPUTING RESEARCHER	2013– 2016
• Developed high-performance parallel graph algorithms, achieving up to 4× speedups on large-scale graph problems.	
NVIDIA	Bengaluru, India
SENIOR SOFTWARE ENGINEER	2012– 2013
• Developed operating system kernel-level drivers for NVIDIA Tegra systems deployed in BMW automotive platforms.	

Efficient LLM & Long-Range Information Flow Research

What Does Muon Amplify?	ICLR 2027 - First author - Under review
Characterized when spectral-reweighting optimizers such as Muon amplify useful rather than noise-dominated gradient directions during LLM pretraining, validating the analysis across Pythia models up to 1B parameters.	
Graph Cascades: Mesoscopic Rewiring Improves Sparse Graph Attention	NeurIPS 2026 - First author
Designed structured sparse-attention topologies for Graph Transformers that improve long-range information flow without dense attention, scaling to graphs with millions of nodes while improving predictive accuracy by up to 11%.	
Schur Preconditioning for Long-Range Message Passing in GNNs	ICLR 2027 - Senior author - Under review
Applied preconditioning ideas from numerical linear algebra to mitigate over-squashing and improve long-range information flow in sparse neural architectures while approximately preserving the original interaction geometry.	
KV-Cache Compression in LLMs by Elimination, Not Eviction	Manuscript 2026 - Sole author
Recasts KV-cache compression as structured elimination that preserves interactions through removed tokens, retaining task-critical tokens in 88–100% of stress tests versus 12–50% for one-shot methods such as Top-K.	

Graph Learning & Numerical Algorithms Research

Canonical Schur Trees for Approximate Cholesky in Laplacian Systems	KDD 2027 - First author - Under review
Developed randomized sparse Schur-tree preconditioners for Laplacian systems underlying large-scale graph learning systems achieving 4.4–4.7× speedups over state-of-the-art solvers.	
Thresholded Local Hyperflow Diffusion	NeurIPS 2026 - First author - Under review
Developed thresholded local hypergraph diffusion with exact local updates, improving clustering on noisy data while reducing active graph volume.	
Fast Local Graph Clustering under Noisy Data	KDD 2027 - Senior author - Under review
Developed side-information-aware local graph clustering via shortest-path search, improving F1 by 6.4 points.	
Adjacency Search Embeddings	TMLR 2025 - First author
Developed node embeddings capturing higher-order structure, improving classification by up to 12.1% in GNNs.	

Education

ETH Zürich, Switzerland	2020– 2024
PhD NETWORK SCIENCE	
Dissertation: Influence Processes on Networks	
Purdue University, USA	2016– 2019
MSc COMPUTER SCIENCE — Focus: Numerical Optimization	

Academic Service & Recognition

☐ Awards:	Wallenberg AI Postdoctoral Fellowship Best Paper Award, HiPC Purdue–Qatar Fellowship AICTE Fellowship
☐ Invited Talks:	Adjacency Search Embeddings, TU Wien, 2025 Influence Processes on Social Networks, UIUC, 2025
☐ Academic Service:	Program Committee: AAAI Peer Review: NeurIPS, ICML, AISTATS